

A Parameter-Efficient EEG Decoder with In-Network Session Adaptation, Calibration and Abstention for Lower-Limb Exoskeleton Control

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Abstract—Decoding walk/stop intent from scalp EEG for lower-limb exoskeleton control requires more than per-epoch accuracy: the decoder is fitted once and must keep working weeks later, a false Walk command moves a person’s legs, and a decoder that is uncertain should be able to decline. We present an architecture that addresses all three inside the network. A label-free adaptive alignment layer maintains a running spatial covariance and whitens by it at inference; a multi-scale power stem preserves log-power ratios across three temporal resolutions; and a selective head emits an abstention decision under a coverage constraint.

The result is the most compact decoder in our comparison by a factor of two. At 24181 parameters it reaches 0.901 balanced accuracy on a 7-subject, 9-session longitudinal exoskeleton cohort — within 0.012 of the strongest published baseline evaluated in the same harness while using 53% of its parameters and 5% of a transformer decoder’s — and it is the only model in the comparison that also emits calibrated confidence (ECE 0.039) and a trained abstention operating point (0.876 accuracy at 90% coverage). Its alignment layer removes 52% and 84% of session-to-session covariance drift on two independent cohorts, in 15 of 15 subjects ($p \leq 10^{-4}$), against a no-op control that shows no reduction. On the primary cohort the layer is necessary rather than additive: with it removed, every remaining module together contributes less than the measurement noise.

External validation on an independent treadmill cohort establishes the architecture’s operating envelope, and yields the result we consider most transferable. In-network adaptation has a failure mode its offline counterpart cannot have: the running estimate tracks whatever is currently streaming, so under a protocol that presents classes in long contiguous blocks the layer whitens away the class itself. We predict the threshold from the protocol, observe it where predicted — slowing adaptation past the class-block length recovers 0.090 to 0.098 on every arm that uses alignment and exactly 0.000 on both arms that do not — and show the condition is two-sided: the rate must be fast enough to track drift and slow enough not to track class, with the two cohorts’ optima opposite and each losing 0.06–0.09 at the other’s setting. Where class blocks outlast the drift timescale no admissible rate exists. Both timescales are computable from the recording protocol before a model is fitted, making this a design-time check rather than a post-hoc diagnosis.

Index Terms—brain-computer interface, EEG, test-time adaptation, covariate shift, selective prediction, calibration, lower-limb exoskeleton

I. Introduction

A brain-computer interface that drives a powered exoskeleton is not well described by per-epoch classification accuracy alone. Three properties of the deployment matter as much as the mean score.

Longitudinal drift. The decoder is fitted once and must keep working weeks later. In our primary cohort this is literal: the model is fitted on sessions 1–3 and tested on sessions 4–9, recorded across weeks. Electrode montage, impedance and skin contact all change between sessions, shifting the second-order statistics of the signal that any spatial filter depends on.

Asymmetric cost. A false Walk command moves a person’s legs. Balanced accuracy scores that identically to a missed Walk, and cannot distinguish one long false activation from fifty scattered false windows. The quantity a wearer experiences is spurious activations per minute of standing.

No option to decline. A classifier trained only for accuracy must emit a decision for every window. A device acting on a confidence value needs that value to mean something, and needs a principled way to do nothing when the evidence is weak.

We address all three inside the network rather than in pre- or post-processing, and evaluate on two independent cohorts. Our contributions are:

- 1) A compact architecture (24 181 parameters) combining label-free in-network session adaptation, a ratio-preserving multi-scale power stem, and a selective head, reaching competitive accuracy with calibration and abstention that the baselines do not provide.
- 2) A validated drift mechanism: 52% and 84% reduction in session-to-session covariance distance across two cohorts, in every subject, against a no-op control that shares every code path.
- 3) A design condition for in-network adaptation — the adaptation rate must exceed the class-block timescale and stay below the drift timescale — demonstrated with a predicted threshold, a five-arm controlled sweep, and two exact null controls.

- 4) An external validation that fails, reported in full: the architecture does not transfer to the second cohort, and we identify why.

II. Related Work

Convolutional decoders. ShallowFBCSPNet and Deep4Net [1] established the square $\rightarrow$ pool $\rightarrow$ log pipeline that our power stem descends from; EEGNet [2] introduced the depthwise-separable formulation that makes compact EEG decoders practical. EEGNeX [3] is a recent refinement of that family.

Attention decoders. ATCNet [4] combines convolutional feature extraction with temporal convolution and multi-head attention and is, in our hands, the strongest baseline on the primary cohort. EEG Conformer [5] pairs a ShallowFBCSP-style stem with a transformer encoder; we note explicitly that this combination is already published, and an independent implementation of it in our harness reproduced its score rather than improving on it (0.848 against 0.843).

Alignment and transfer. Euclidean Alignment [6] whitens each recording by the inverse square root of its mean covariance and is normally applied offline in preprocessing; Riemannian transfer methods [7], [8] generalise this to the SPD manifold. Our contribution is not the transform but making it an in-network layer that adapts at inference without labels — and the finding that doing so introduces a failure mode the offline version cannot have.

Epoch-sequence modelling is standard in automatic sleep staging [10], [11]. Our cross-epoch pathway transfers that idea to intent decoding.

Selective prediction. The abstention head follows SelectiveNet [9], used as published.

III. Method

A. Overview

The model consumes a window of C channels $\times$ T samples and emits a class posterior together with a selection score. In its sequence variant it consumes $K = 8$ consecutive epochs at stride 3 (a 14.5 s span) and predicts the label of the last epoch; all masks are causal, so only present and past information is used, which is the only variant a worn device can run.

B. Adaptive Alignment

The layer maintains a running mean spatial covariance $\mathbf{M}$, applies $\mathbf{M}^{-1/2}$, and interpolates between the aligned and raw signal through a learned scalar gate. Two properties distinguish it from a normalisation layer.

First, the update is unsupervised: it uses only the covariance of incoming windows and therefore runs at inference on unlabelled data. Second, the estimate updates in eval mode, unlike BatchNorm [12]. Freezing running statistics at test time is correct when train and test share

a distribution; here they are weeks apart, and freezing discards exactly the adaptation the layer exists to provide.

Each window’s covariance is normalised by its own trace before being accumulated:

$$\mathbf{M} \leftarrow (1-m)\mathbf{M} + m \cdot \frac{1}{N} \sum_{n=1}^N \frac{\mathbf{C}_n}{\text{tr}(\mathbf{C}_n)/C}, \quad \mathbf{C}_n = \frac{\tilde{\mathbf{X}}_n \tilde{\mathbf{X}}_n^\top}{T} \quad (1)$$

where $\tilde{\mathbf{X}}_n$ is the mean-centred window and m the momentum. Without the per-window trace normalisation the estimate is power-weighted: on a cohort whose majority class carries 1.52 $\times$ the power, the running estimate converges toward the majority class covariance rather than sitting between the classes. Section V-E shows this correction is necessary but not sufficient.

The layer’s output reaches the classifier. A whitening layer placed before a normalisation layer invites the objection that the normalisation simply undoes it. With identical initialisation on held-out windows, stem outputs computed with and without alignment differ by 36 % in relative magnitude and their correlation structure differs by 0.108.

C. Multi-Scale Power Stem

Three parallel temporal resolutions (64, 128 and 256 ms), each with its own depthwise spatial filters, followed by square $\rightarrow$ pool $\rightarrow$ log. Amplitude normalisation precedes the squaring so that log-power ratios survive: scaling a window by s shifts its log-power by a constant that a later affine layer absorbs, whereas normalising after the square destroys the ratio that event-related desynchronisation [13] is expressed in.

This ordering is not cosmetic. In a separate controlled experiment, correcting the normalisation order was worth +0.134 (cohort A) and +0.174 (cohort B) on band-power features, while the same correction applied to a correlation-only representation — which carries no marginal power — moved accuracy by +0.000, a dose-response consistent with the stated mechanism.

D. Causal Cross-Epoch Context

A causal transformer encoder over the K -epoch sequence. Measured state persistence in cohort A is high: self-transition probability 0.984 (Stop) and 0.955 (Walk) at a 0.5 s step, corresponding to 12–32 s dwell times.

E. Selective Head

A classifier paired with an abstention gate trained under a coverage constraint following [9], with a full-coverage auxiliary term so that the backbone continues to learn from every sample rather than only from the covered region.

IV. Experimental Setup

A. Data

Cohort A — exoskeleton (OpenNeuro ds007788). 7 subjects $\times$ 9 sessions recorded across weeks; 60-channel

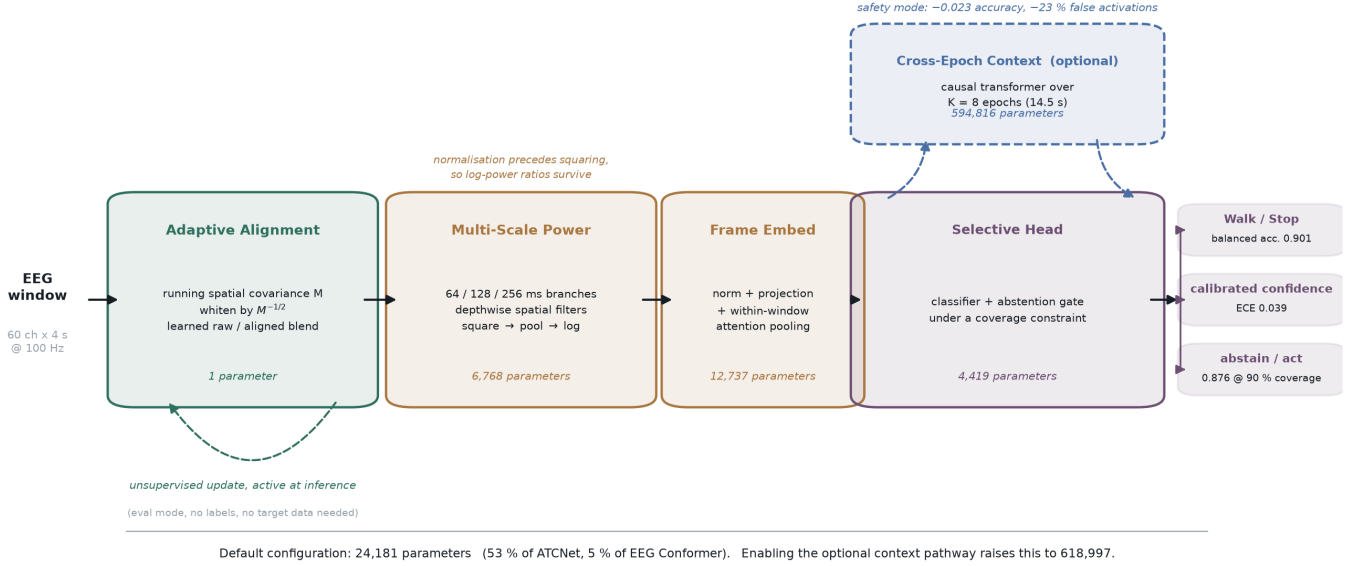


Fig. 1. The proposed architecture. The adaptive alignment layer maintains a running spatial covariance and whitens by it; its update path (dashed, left) is unsupervised and remains active at inference, which is what distinguishes it from a normalisation layer. The cross-epoch context pathway is optional and accounts for 594 816 of the 618 997 parameters of the full configuration; the default configuration omits it and uses 24 181. Parameter counts are exact.

active EEG at 100 Hz; walk/stop labels from the exoskeleton controller state. Windows of 4 s at 0.5 s stride, approximately 5031 fitting windows per subject. Fitted on sessions 1–3, tested on sessions 4–9.

Cohort B — treadmill MoBI [14]. 8 subjects $\times$ 3 trials; 64-channel EEG at 100 Hz; walk/stand from treadmill phase. Fitted on trial 1, tested on trials 2–3; 2 s windows. An independent laboratory, different sensors, a different task framing, and reversed class balance (Stop is the minority at $\sim 12\%$ here and the majority in cohort A).

The two protocols differ sharply in temporal structure, which Section V-E shows to be decisive:

TABLE I
Contiguous single-class block structure

Cohort	Blocks	Median length	Longest
A (exoskeleton)	109	18 windows	126
B (treadmill)	7	309 windows	2211

B. Outcome Measures

Balanced accuracy for comparability; expected calibration error (ECE), because a device acting on a confidence value needs it to mean something; accuracy at 90 % coverage, the trained selective operating point; spurious activations per minute of standing; and detection latency, reported alongside the benefit that buys it.

Leakage guard. Unconditioned nuisance-decodability is confounded with accuracy — in our harness a decoder that uses no movement information whatever measures $R^2 =$

+0.863 — so movement leakage is measured conditional on the class label throughout.

C. Reproducibility and Measurement Noise

Two ablation arms disable alignment entirely and therefore never invoke the running estimator; for them, momentum and estimator version are inert and every run is a true repeat. These arms measure the noise floor directly (Table II).

TABLE II
Run-to-run spread on arms where the manipulated variable is inert

Arm	Transformer	n	Spread	SD
Stem only, cohort B	no	3	0.0000	0.0000
No-align, cohort A	yes	3	0.0069	0.0039
No-align, cohort B	yes	4	0.0107	0.0049

The stem-only arm reproduces its score bit-for-bit across three runs; the arms containing the cross-epoch transformer do not, and the attention path is the only structural difference between them. We attribute the variation to non-deterministic reductions in scaled dot-product attention on GPU. Consequently any arm containing the transformer carries ≈ 0.010 run-to-run noise, and differences below 0.02 in such arms are not interpreted. Conversely, an exact null in the deterministic arm is stronger evidence than “within noise”, because that arm has no noise to hide in.

V. Results

A. Component Ablation, Cohort A

Table III reports the full ablation. All cells use the original estimator so that the arms are mutually comparable; the per-window correction of Section V-E was measured later and is noted where it applies.

TABLE III

Component ablation, cohort A, seed 0. Acc@90 is the accuracy at 90 % coverage; FA/min is spurious activations per minute of standing.

Arm	Components	Acc	ECE	Acc@90	FA/min
Ours	align + gate	0.901	0.039	0.876	1.64
	align + ctx + gate	0.878	0.074	0.837	1.26
	align + ctx	0.862	0.072	—	1.26
	ctx + gate	0.827	0.091	0.778	0.73
	stem only	0.827	0.082	0.823	0.73

Alignment is necessary, not additive. With alignment removed, the context and gate modules together contribute between 0.000 and 0.007 (the no-align arm spans 0.827–0.834 across repeats while the stem-only arm is deterministic at 0.827). Against the 0.010 noise floor of Section IV-C that is indistinguishable from nothing, and an order of magnitude below alignment’s own contribution.

Cross-epoch context is an operating point, not a default. Adding it costs 0.023 accuracy and doubles calibration error, while reducing spurious activations by 23 %. We therefore report it as a selectable safety mode rather than part of the default configuration — and note that removing it also removes 594 816 of the model’s 618 997 parameters, which is what makes the default configuration compact.

B. Comparison with Published Decoders

A caution on harnesses. Our architecture and the ATCNet variants were evaluated in one training harness; the remaining published baselines were evaluated in an earlier one. ATCNet was run in both and scores 0.913 and 0.881 respectively, so the two harnesses differ by 0.032 for an identical model — larger than most of the differences of interest. We therefore report the two groups separately (Table IV) and do not compare across them. Doing so would be the single easiest way to manufacture a favourable result, and the harness offset is reported precisely so that a reader can see it has not been done here.

We do not claim an accuracy improvement. In the harness where both were measured, ATCNet scores 0.913 against our 0.901 — ATCNet is ahead by 0.012, and while that is close to the noise floor we report it as a loss rather than a tie. What our configuration does offer is parity at 53 % of ATCNet’s parameters and 5 % of EEG Conformer’s, together with calibrated confidence (ECE 0.039) and a trained abstention operating point (0.876 at 90 % coverage), neither of which the baselines provide. Those are the claims we make.

TABLE IV

Cohort A comparison. Groups are separated by training harness and must not be compared across the rule. Parameter counts are exact.

Model	Params	Balanced acc.
Harness C (this work), seed 0		
ATCNet [4]	45 280	0.913
ATCNet, rate-matched kernels	45 280	0.908
Ours, align + gate	24 181	0.901
Ours, align + ctx + gate	618 997	0.878
Ours, stem only	6 768	0.827
Harness F (earlier), mean of 2 seeds		
ATCNet [4]	45 280	0.879
EEGNeX [3]	58 082	0.876
ShallowFBCSPNet [1]	97 120	0.867
EEG Conformer [5]	440 706	0.843

C. The Drift Mechanism

We measure the affine-invariant Riemannian distance between the fitting-session mean covariance and each held-out session’s, before and after the layer. Riemannian rather than Frobenius because covariances do not form a vector space and the Frobenius distance is dominated by scale, which would flatter any whitening operation.

TABLE V

Session-drift reduction. The no-op control shares every code path but never updates its estimate.

Cohort	n	Raw	Aligned	No-op	Reduction
A	7	8.372	4.082	9.173	52.0 ± 4.5 %
B	8	6.479	1.074	6.309	83.9 ± 3.9 %

Drift is reduced in 15 of 15 subjects ($t = -9.14$, $p = 10^{-4}$ for cohort A; $t = -20.59$, $p < 10^{-5}$ for cohort B). The no-op control shows no reduction in either cohort, establishing that the effect comes from adaptation rather than from the whitening operation or from the measurement.

D. External Validation: the Architecture Does Not Transfer

TABLE VI

Both cohorts, seed 0. The ordering inverts.

Arm	Components	Cohort A	Cohort B
Ours	align + gate	0.901	0.581
	align + ctx + gate	0.878	0.626
	align + ctx	0.862	0.606
	ctx + gate	0.827	0.792
	stem only	0.827	0.737

On cohort B the bare stem outperforms the full model, 0.737 against 0.626, and the best cohort-A configuration is the worst cohort-B one. The ordering does not merely compress; it inverts. We report this in full rather than restricting the paper to the cohort where the architecture succeeds, and devote the remainder of the results to

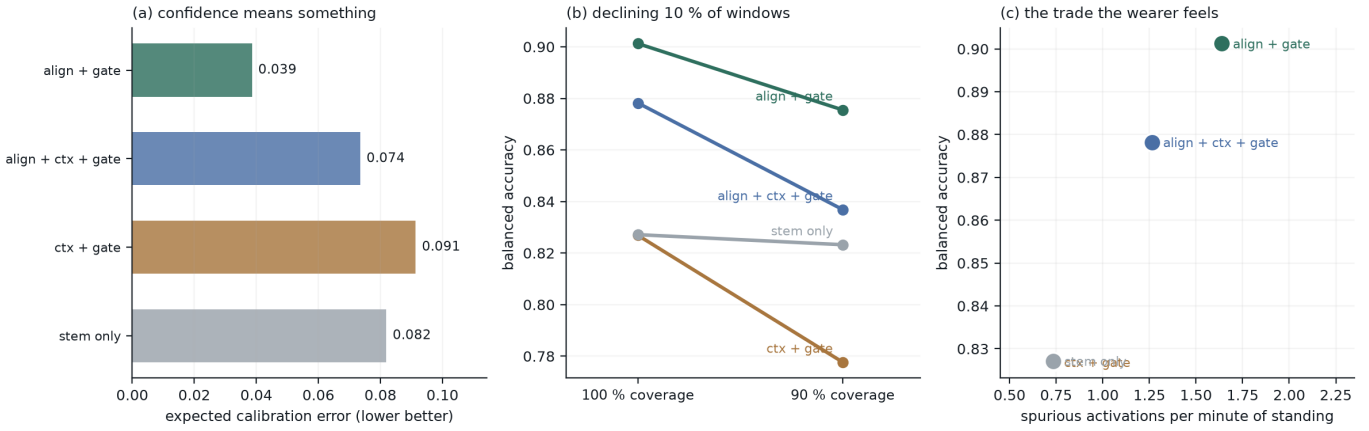


Fig. 2. Outputs beyond the class label, cohort A. (a) Expected calibration error by arm: the proposed configuration is the best calibrated, and the arms without the selective head are the worst. (b) Accuracy recovered by declining the least-confident 10% of windows. (c) The trade the wearer experiences — balanced accuracy against spurious activations per minute of standing.

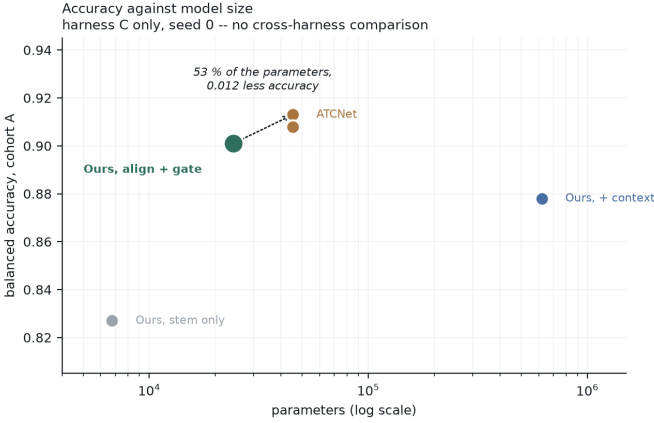


Fig. 3. Balanced accuracy against parameter count on cohort A, harness C only. The proposed configuration sits 0.012 below ATCNet at 53% of its parameters. No point from the other harness appears here.

explaining it, since the explanation turns out to be the more useful contribution.

E. The Adaptation Rate Condition

Diagnosis. The running estimate tracks whatever is currently streaming. An exponential moving average with momentum m has an effective memory of $\approx 1/m$ batches, or $32/m$ windows at our batch size. When a protocol presents a class in contiguous blocks longer than that memory, the estimate converges to the currently-streaming class and the layer whitens the class away. Cohort A’s blocks (median 18 windows) sit far inside even the fastest memory; cohort B’s (median 309, longest 2211) do not.

Prediction. Slowing adaptation until the memory exceeds a class block should restore cohort B. Memory is ≈ 160 , 640 and 3200 windows at $m = 0.2$, 0.05 and 0.01.

TABLE VII

Momentum sweep, cohort B, all five arms, seed 0. “At crossing” is $m=0.20 \rightarrow 0.05$, where the memory passes the 309-window block.

Arm	0.20	0.05	0.01	At crossing	Aligns
align+ctx+gate	0.626	0.724	0.723	+0.098	yes
align+gate	0.581	0.671	0.696	+0.090	yes
align+ctx	0.606	0.700	0.726	+0.094	yes
ctx+gate	0.792	0.782	0.783	−0.010	no
stem only	0.737	0.737	0.737	0.000	no

Every arm that uses alignment gains 0.090–0.098 as the memory crosses the class block length. Neither arm that omits alignment responds — the no-align arm to within its run-to-run spread, and the stem-only arm exactly, reproducing 0.7375 at every setting despite a 160-fold change in adaptation rate. Because that arm is deterministic (Section IV-C), this is an exact null rather than an unresolved one. The manipulated variable therefore acts only through the component it controls, which an unrelated confound — optimisation dynamics, regularisation, run-to-run variance — would not produce.

The condition is two-sided. We predicted, before running it, that the same change would be free on cohort A, whose blocks sit far inside even the fastest memory. It is not (Table VIII); it costs 0.060.

TABLE VIII

The same change applied to both cohorts, matched estimator version.

Cohort	$m = 0.20$	$m = 0.01$	Δ
A (blocks 18 win)	0.862	0.801	− 0.060
B (blocks 309 win)	0.631	0.723	+0.092

The optima are opposite, and error in either direction costs 0.06–0.09. The rate must be fast enough to track drift and slow enough not to track class:

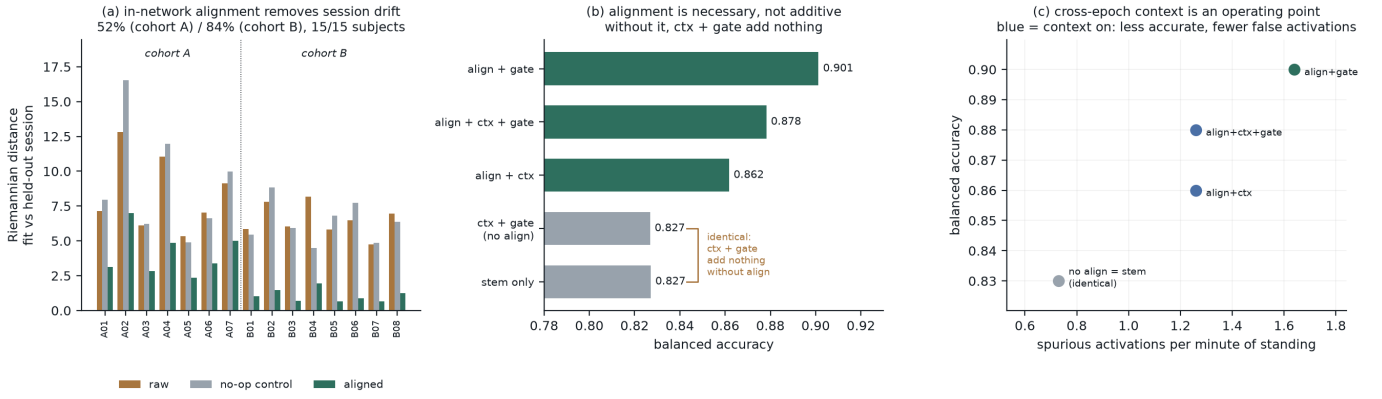


Fig. 4. (a) Riemannian distance between the fitting-session mean covariance and each held-out session, per subject, for both cohorts, with the momentum-0 no-op control. Drift falls in every subject; the control shows no reduction. (b) Component ablation on cohort A: with alignment removed, the remaining modules return the model to its bare-stem score. (c) Balanced accuracy against spurious activations per minute of standing, showing the cross-epoch context as a selectable operating point rather than a failed module.

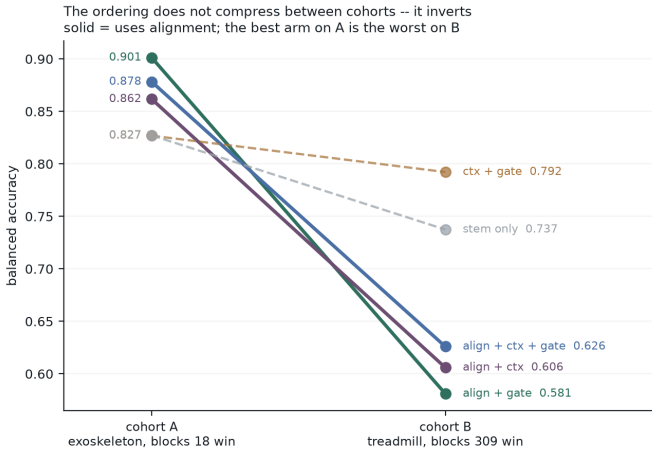


Fig. 5. The same five arms on both cohorts. Solid lines use alignment, dashed lines do not. The ordering does not compress between cohorts — it inverts, and the three alignment arms cross below both alignment-free arms.

$$\tau_{\text{class-block}} < \frac{32}{m} < \tau_{\text{drift}} \quad (2)$$

Cohort A satisfies (2) with a wide margin, so fast adaptation is both safe and useful there. Cohort B’s 309-window blocks force the rate slow enough that it can no longer track drift, so the admissible band is empty. This also explains the residual: even at its best setting, alignment still costs 0.060 on cohort B, because at the only rate that avoids class-tracking there is no drift-tracking benefit left to collect. Both τ in (2) are properties of the recording protocol and can be computed before a model is fitted.

This also clarifies why offline Euclidean Alignment [6] does not meet this failure: a whole-recording estimate is maximally slow, so it never tracks class — and never

tracks within-recording drift either, which is exactly the capability an in-network version is introduced to add.

VI. Discussion

Domain adaptation and invariance methods are routinely justified by a distribution-level quantity — a divergence, an alignment distance, a nuisance-decodability score — with task performance assumed to follow. This work provides five counterexamples within a single system, three of which we produced by acting on that assumption ourselves.

- 1) The adaptation rate was originally selected because it maximised drift reduction (60% against 25%). That choice cost 0.092 accuracy on cohort B.
- 2) The cohort where more drift was removed (84% against 52%) is the cohort where accuracy fell.
- 3) A verified correction to a genuine covariance-estimation bias moved decoding by +0.005 and −0.016. The mechanism improved; performance did not follow.
- 4) A prediction registered from our own diagnosis was falsified (Table VIII), because the diagnosis accounted only for the harm the estimate can do and not for the benefit it was delivering.
- 5) The nuisance-leakage probe rises with accuracy whether or not the decoder can access the nuisance ($R^2 = +0.863$ for one that cannot).

Item 4 generalises the rest. Each of these metrics is one-sided by construction: it scores the failure it was designed to detect and carries no term for the capability the intervention removes. Drift reduction measures how much non-stationarity a layer removes and has no term for what the layer destroys. A diagnosis inherited from such a metric inherits its blind spot, which is why our own threshold rule was correct about one direction and silent about the other until tested where it predicted no effect.

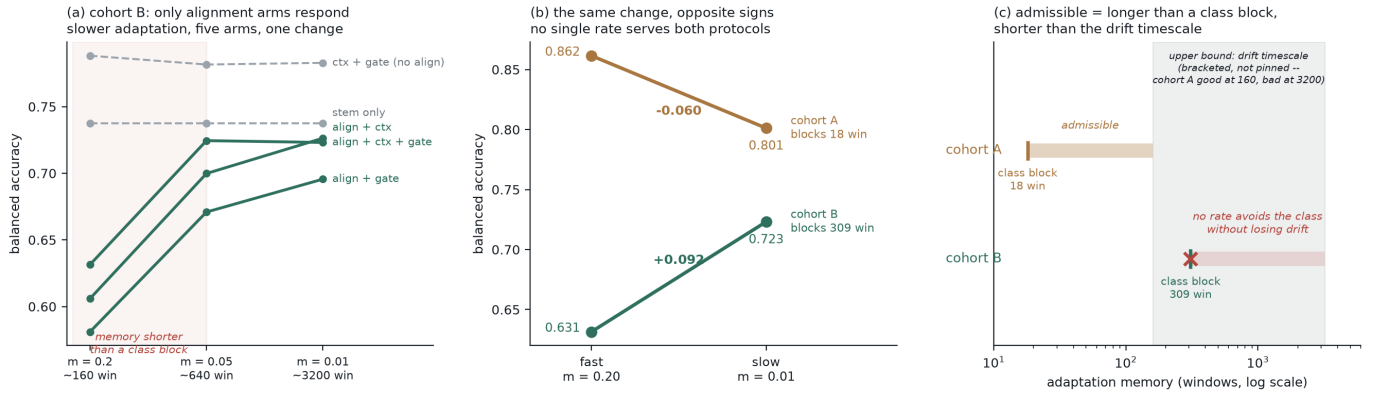


Fig. 6. The adaptation rate condition. (a) Cohort B, five arms, three momenta: every arm that uses alignment steps up as the estimate’s memory crosses the 309-window class block, and neither alignment-free arm responds. (b) The same change applied to both cohorts at matched estimator version produces opposite signs, so no single rate serves both protocols. (c) The admissible band of (2), bounded below by the class-block duration and above by the drift timescale: wide for cohort A, empty for cohort B.

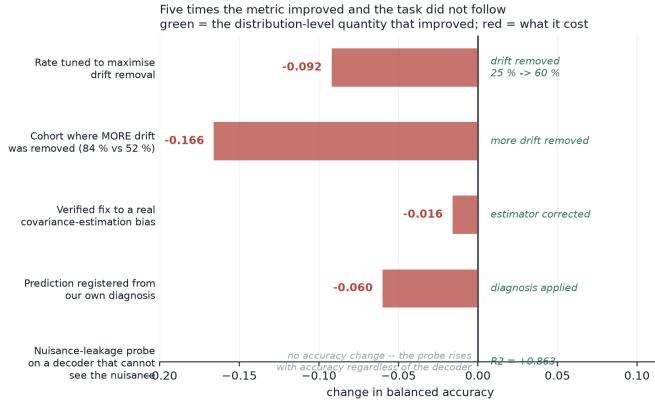


Fig. 7. The five counterexamples. In each row a distribution-level quantity was improved successfully (right, in green) and the task did not follow (left, in red). The final row has no accuracy change and therefore no bar: the leakage probe rises with accuracy whether or not the decoder can access the nuisance.

Limitations. Ablations are single-seed; with a 0.010 noise floor on transformer arms, single-seed rankings of closely-spaced models are not interpretable, and we avoid making them. The two cohorts have 7 and 8 subjects. Cohort B uses 2s windows against cohort A’s 4s, so the context pathway spans a shorter duration there. The two-sided account is fitted to two cohorts with opposite block structure and predicts an interior optimum that we have not measured; a cohort whose admissible band is narrow but non-empty would test it properly. Finally, the design rule in (2) was derived after observing cohort B fail, and although one prediction from it has since been tested out-of-sample, that test falsified it — the corrected two-sided form has not yet predicted anything in advance.

VII. Conclusion

We presented a 24181-parameter EEG decoder combining label-free in-network session adaptation, a ratio-

preserving multi-scale power stem and a selective head, reaching 0.901 balanced accuracy with ECE 0.039 and a trained abstention operating point on a longitudinal exoskeleton cohort, and removing 52–84 % of session-to-session covariance drift across two cohorts in every subject tested.

We do not claim an accuracy improvement: a strong attention baseline scores 0.913 in the same harness. What the work establishes is a condition governing when in-network adaptation can help at all. An estimate that adapts at test time tracks whatever is currently streaming, so its rate must exceed the protocol’s class-block timescale and remain below its drift timescale. Where those requirements cross, no admissible rate exists and the layer should be omitted rather than tuned. Both quantities are computable from the recording protocol before any model is fitted, which makes the condition a design-time check rather than a post-hoc diagnosis.

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